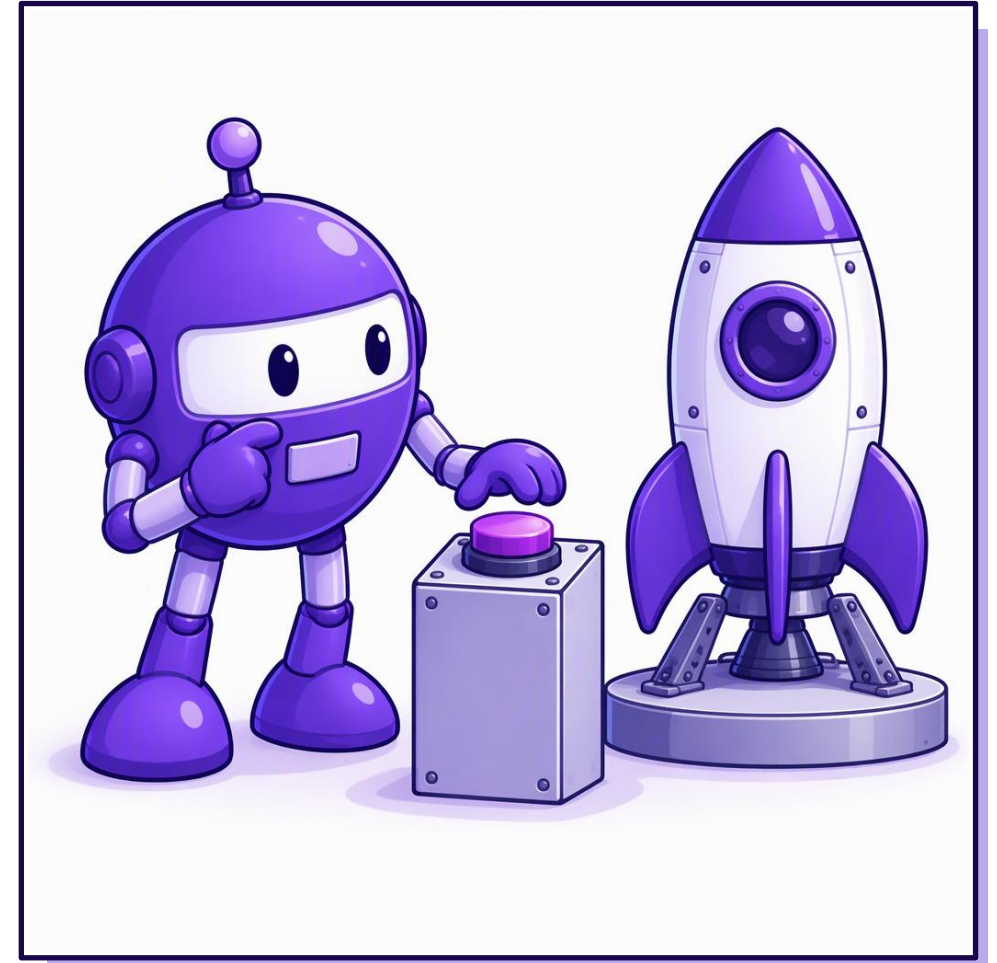




It works. Can we trust it?

From a clever demo to an agent
you can operate.

ONE AGENT. A BETTER STANDARD OF PROOF.



Production Ready: Observability, Governance and Deployment

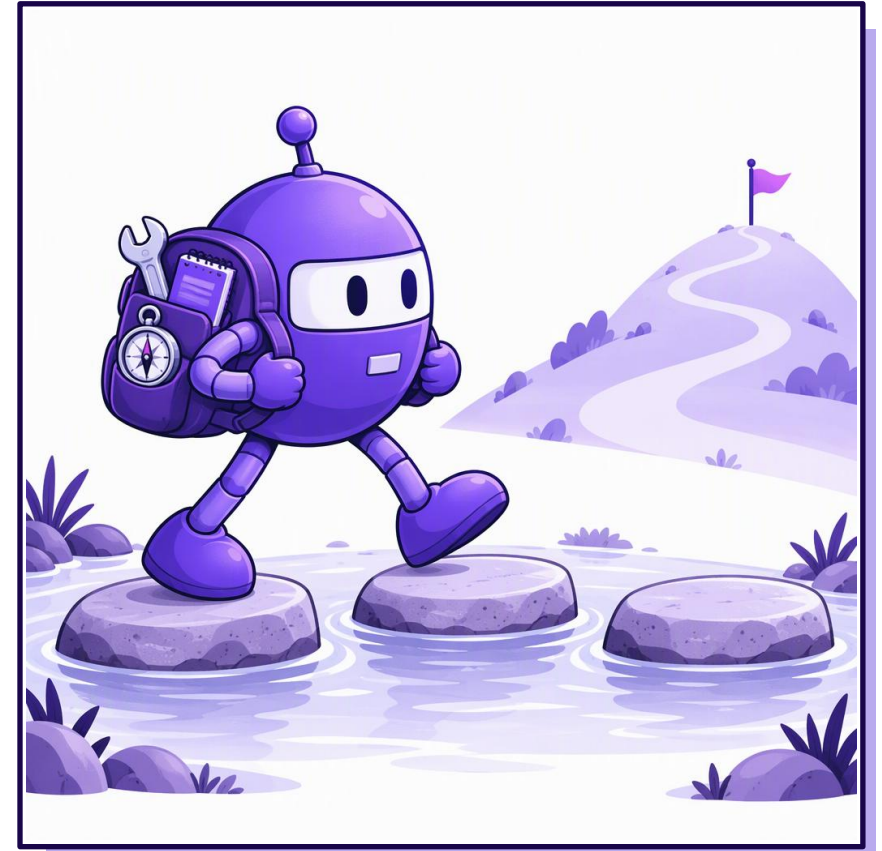


Three sessions. A bigger backpack.

01 Tools + planning

02 Files + approval + memory

03 Skills + bounded execution



More capability brings more responsibility.



Three questions before we ship.

Connect

50 / 51

Can it use external tools?

Resume

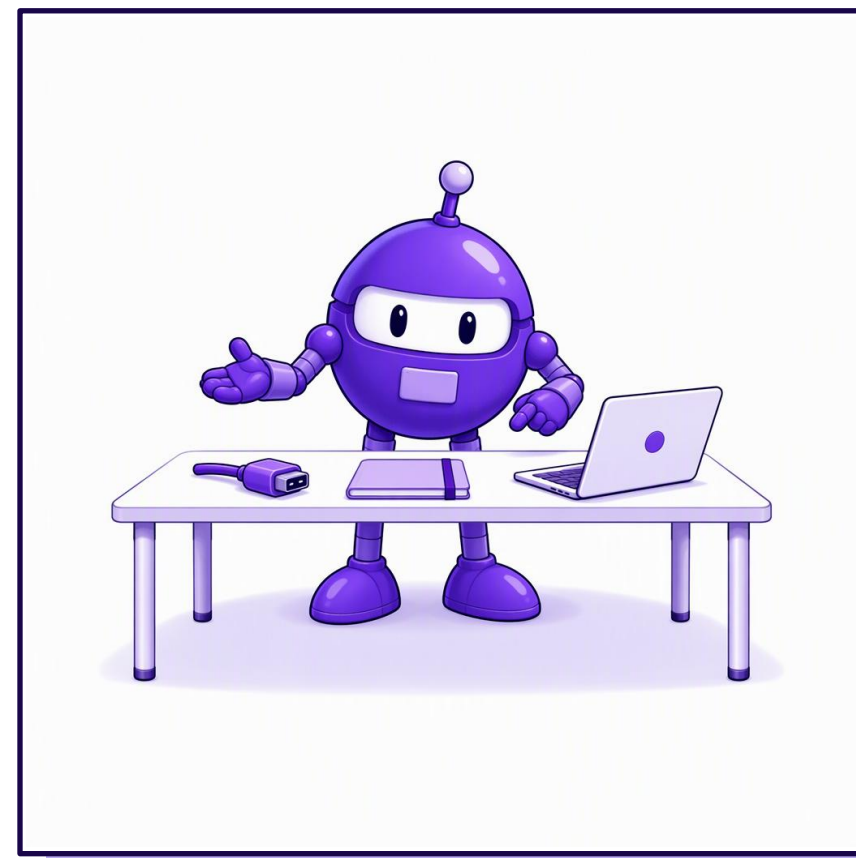
60 / 61

Can it pick up after a restart?

Run locally

70 / 71

Can inference stay on-device?



Same teaching pattern: plain C# first, then the MAF bridge.



One question. Two tool routes.

What's new in .NET 10?

WebSearchAgent

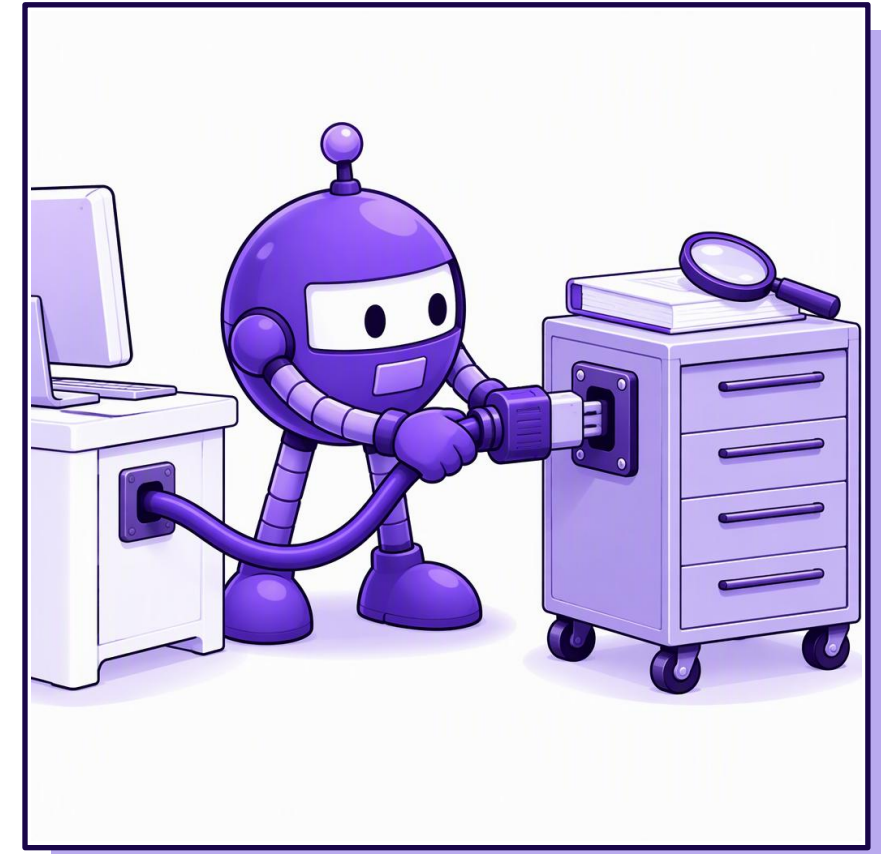
Hosted web search

Built-in tool; not MCP

MicrosoftLearnAgent

Microsoft Learn MCP

External tools via the harness



MCP connects tools. The harness still controls their use.



A restart should not end the story.



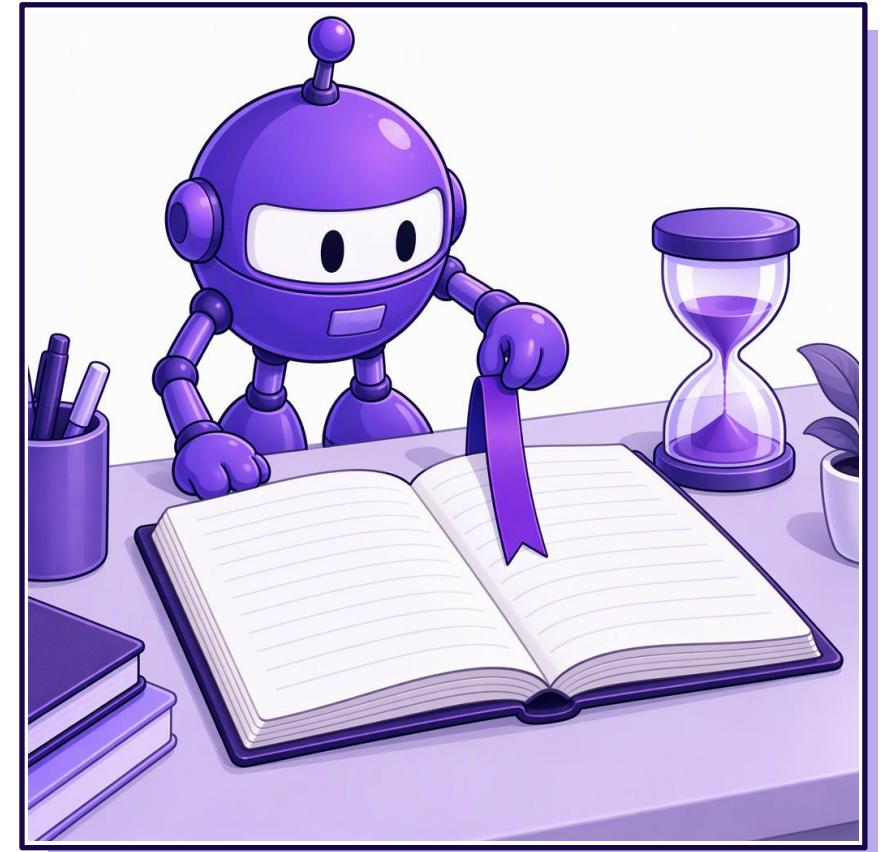
60 / Plain C#

Explicit transcript JSON

61 / MAF

Serialize + restore AgentSession

Proof: prior turns reach the model again.



Session state resumes this conversation. Memory spans conversations.



Same harness. Model on your machine.

70 / Plain C#

OpenAI SDK -> local endpoint

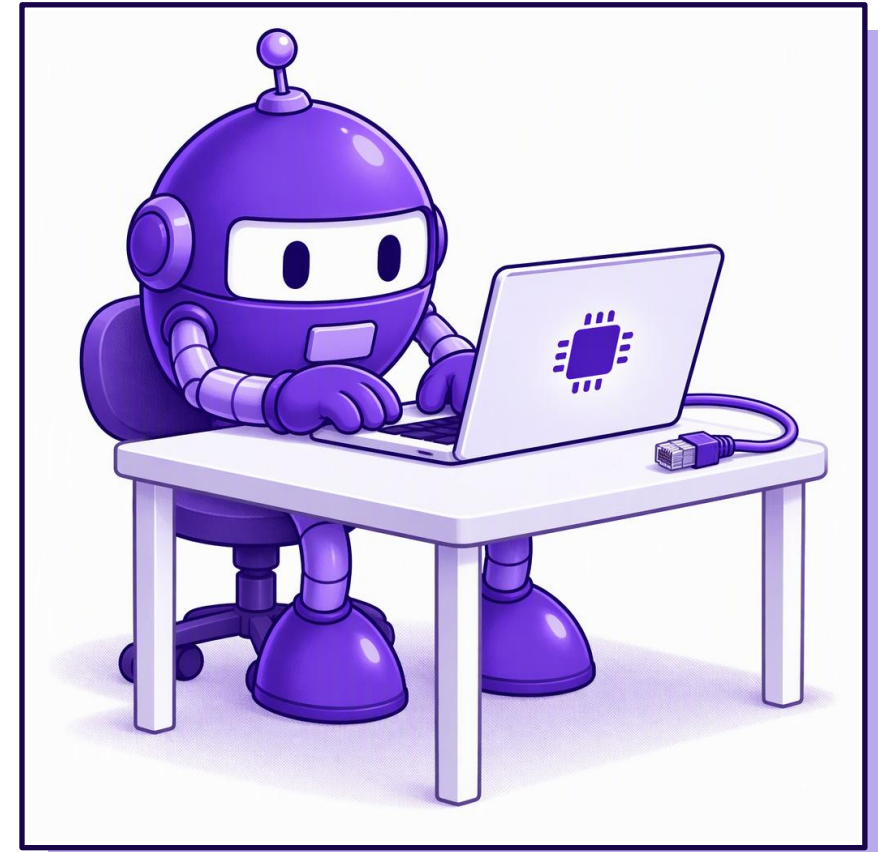
71 / MAF

IClient -> AsHarnessAgent

MODEL DOWNLOAD / SETUP FIRST

On-device inference with phi-4-mini

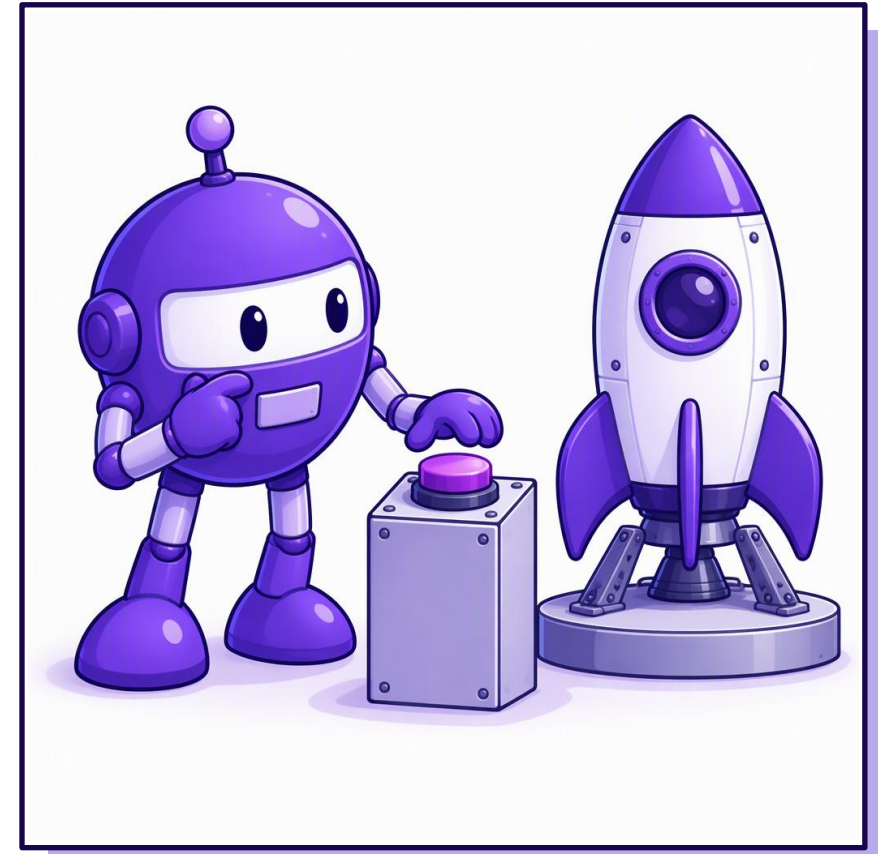
Pre-cache the model. Prove a real tool call, not just prose.





"It answered" is not enough.

- ? Did the tool actually run?
- ? Was the action allowed?
- ? Will the next version regress?



A plausible answer is a claim, not execution evidence.



Earn trust in four steps.

01

Observe

What
happened?

02

Govern

Was it
allowed?

03

Evaluate

Did anything
regress?

04

Host

Where can it
run?

Observe -> govern -> evaluate -> host.



Follow the request, not the guess.



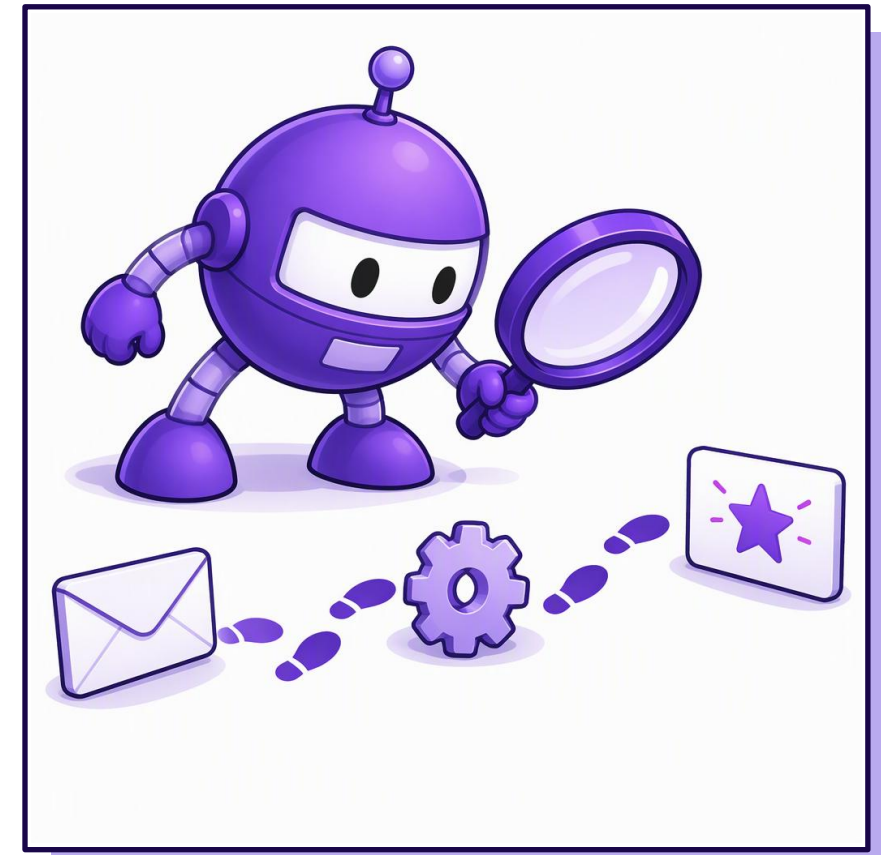
ActivitySource

What happened in this request

Meter

How often; how long

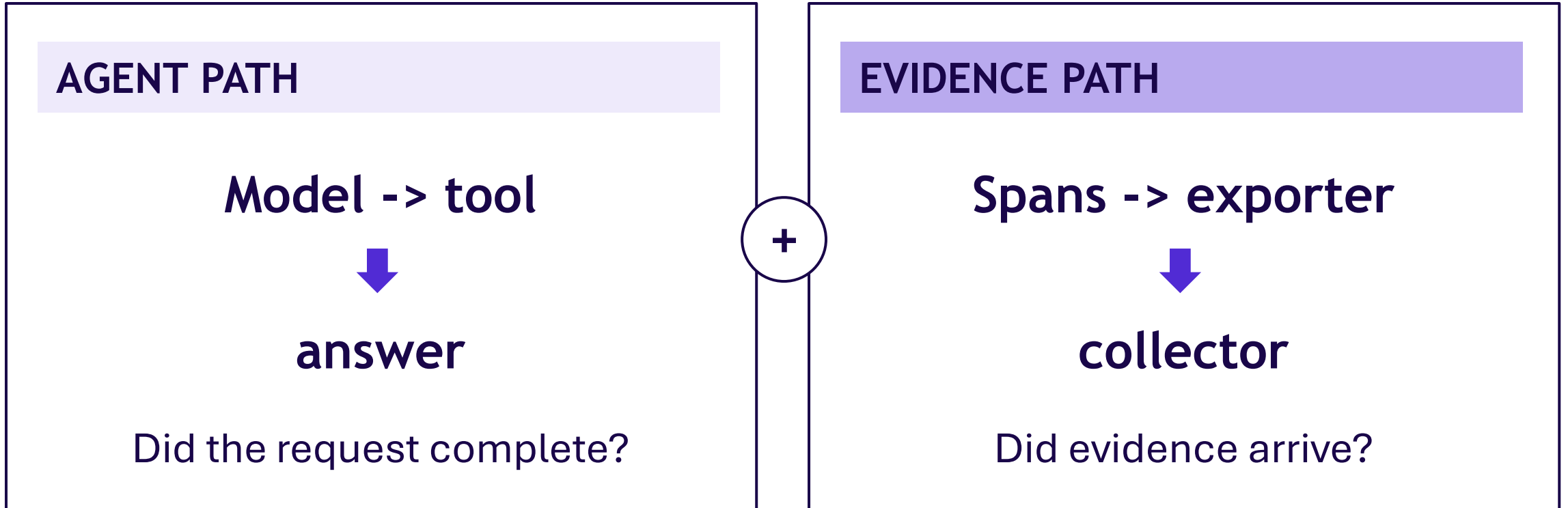
CONCEPTUAL TRACE / NOT MEASURED TIMINGS



A trace links cause and effect. A metric summarizes behavior.



An answer and a trace are two results.



Inference success does not prove telemetry delivery.



Keep the trail. Leave the secrets.

KEEP

Correlation

Duration

Safe outcome

LEAVE OUT

Raw prompts

Tool payloads

Secrets + file contents

Test captured records, including failure paths.



Permission is not one checkbox.

Authorization

May this profile do it?

Human approval

Do you consent to this action?

Data policy

May this content pass?



Approval cannot override a forbidden action.



"No" must mean nothing happened.



0

simulated trades
after denial



EXPECTED ASSERTION / NOT A CAPTURE

Denied, EOF, timeout or cancellation: no side effect.



Unavailable is not a policy pass.

ALLOWED

Proceed within policy

BLOCKED

Stop before disclosure

UNAVAILABLE

Stop; report the service gate

Purview: entitlement + configuration + identity + authorization

Purview enabled but unavailable? Protected operations fail closed.



Name both the runner and the grader.

30 / Plain C#

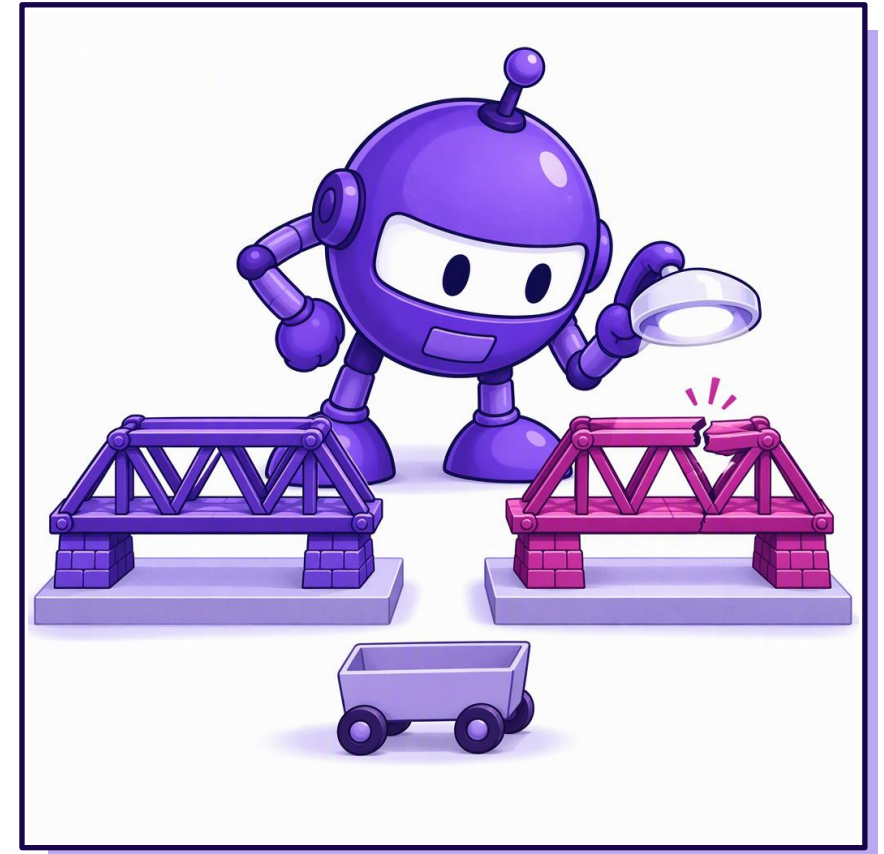
Deterministic local assertions

31 / MAF

Fixture or live inference; local grading

32 / Optional

Separate Foundry service grading



Fixture inference is not live inference. Remote grading is a separate gate.



Break it on purpose. Watch it fail.



Expected sequence, not a recording

Good -> bad -> good. Preserve all three outcomes.



Ship less power. Keep the purpose.

LOCAL

Scoped files + memory
Approved shell + CodeAct

HOSTED

No local file tools or memory
No shell or generated code



A working directory is not a sandbox.



HTTP 200 is not the finish line.



HTTP 200

HTTP transport succeeded

completed

Agent operation completed

Assert response.status == completed.



A bundle is not a deployment.

PACKAGE

Complete .NET output + assets

AUTHORIZE

Identity + access + cost

DEPLOY

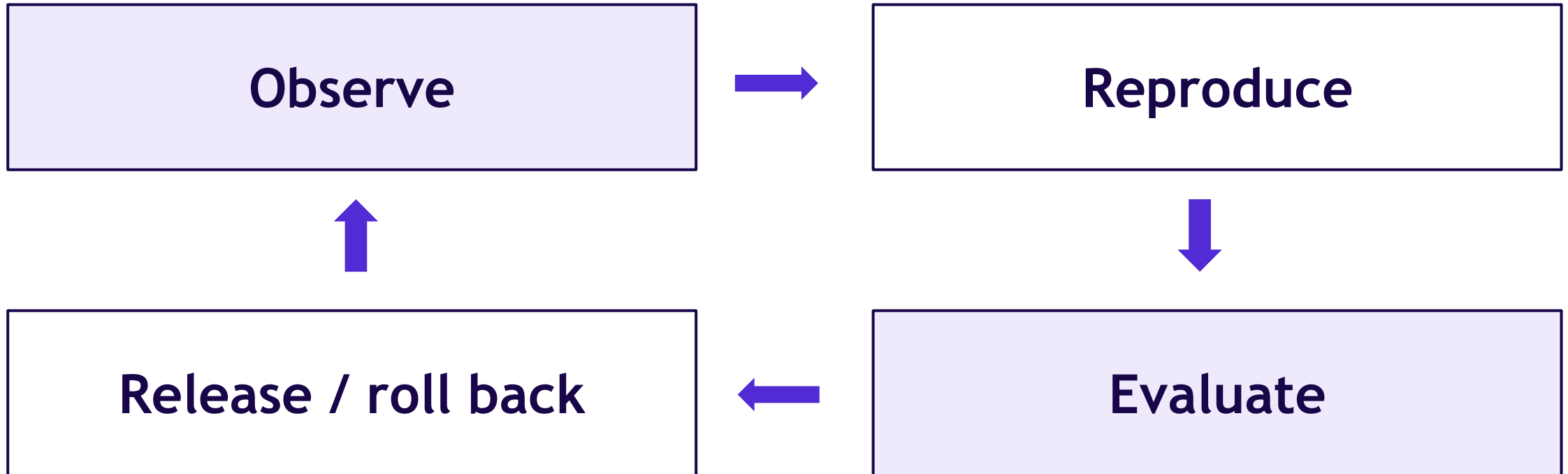
Only after the explicit gate



Build evidence, deployment evidence and approval are separate.



Make the next change deliberate.



A rollback plan is not proof that rollback was exercised.



Now bring the story together.

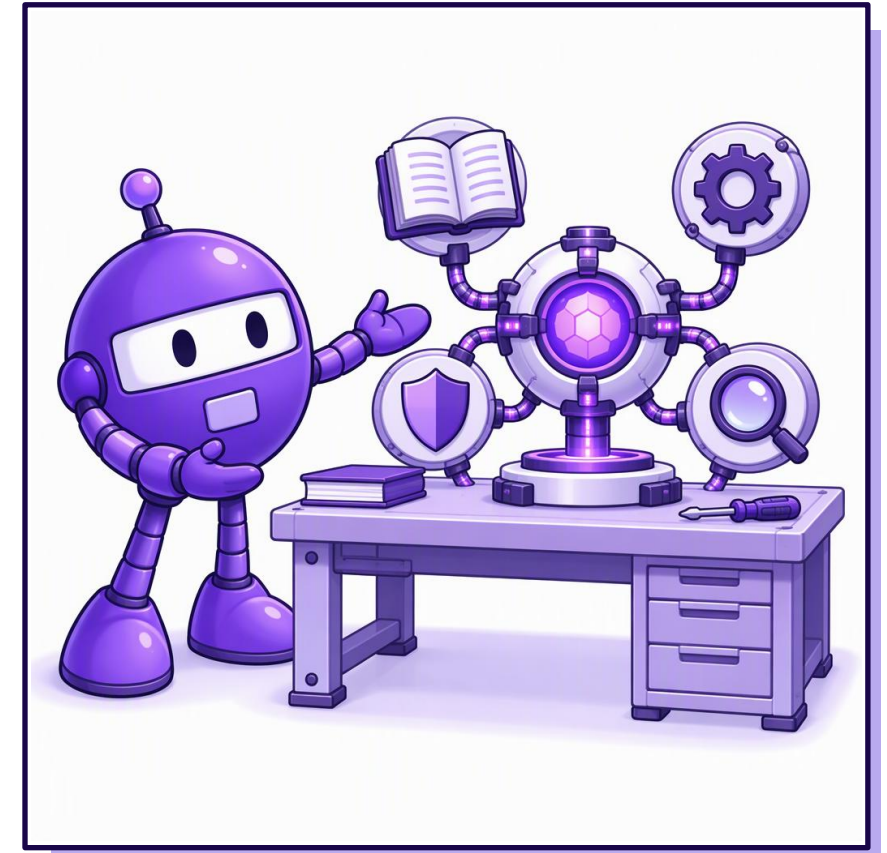
Sessions 1-3

Tools, scoped state, bounded execution

Session 4

Observe, govern, evaluate, host

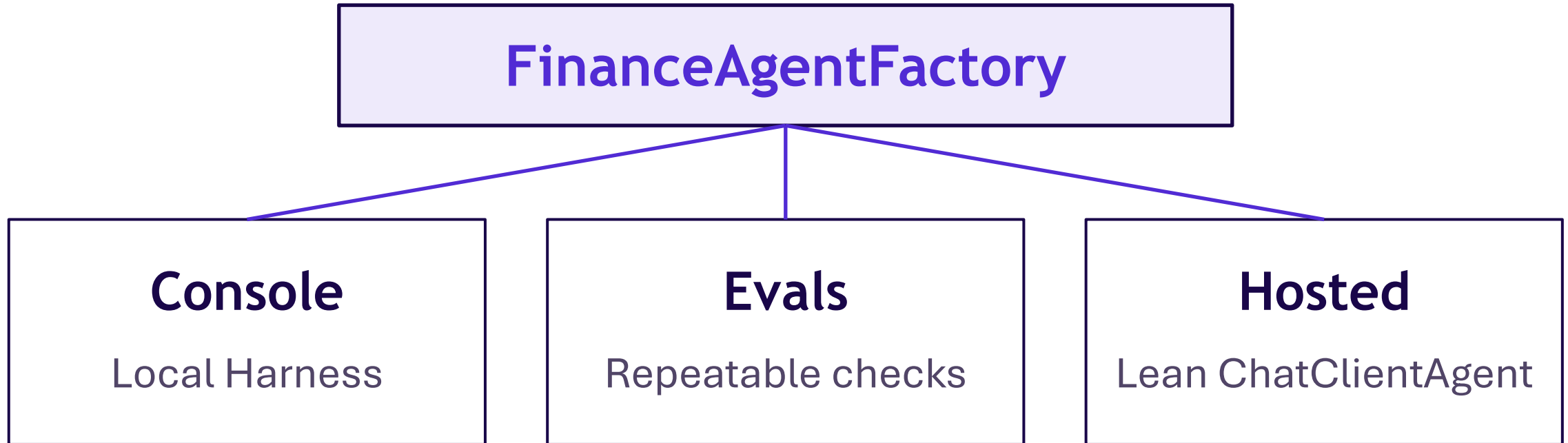
SYNTHETIC DATA / EDUCATIONAL APP



One composed application. Different host authority.



One definition. Three ways to run it.



Separate samples: MCP 50/51, persistence 60/61, Foundry Local 70/71

Same domain definition does not mean identical capabilities or history.

Build capability. Keep the proof.

SAMPLES

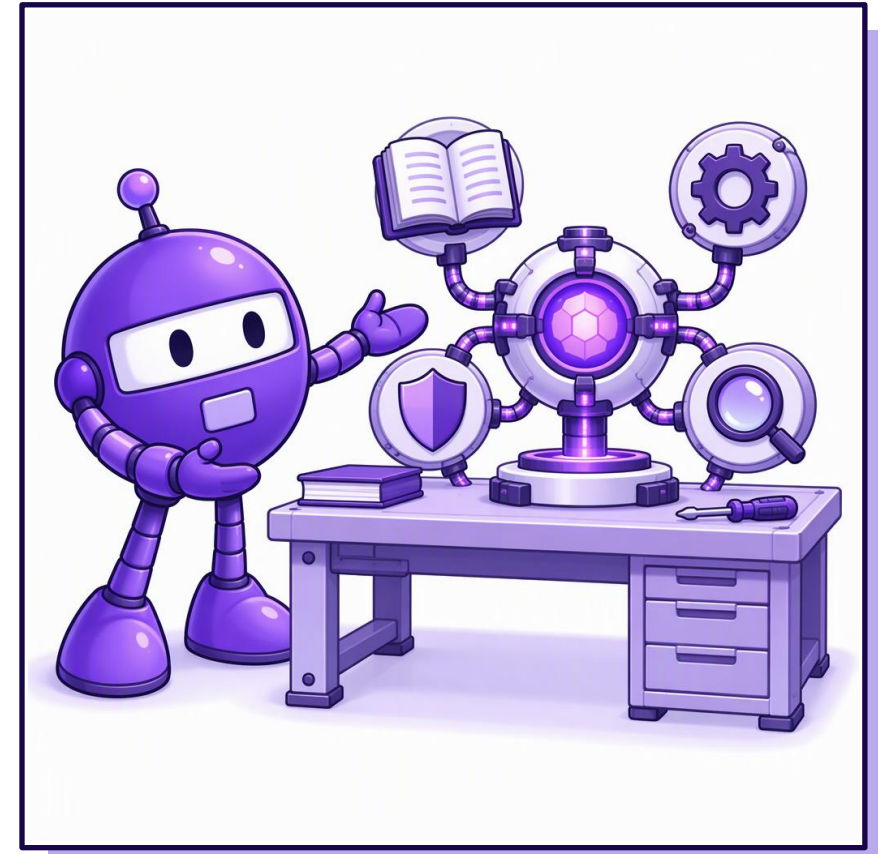
aka.ms/mafclaw/repo

SERIES

aka.ms/mafclaw

SESSION 4

aka.ms/mafclaw/4

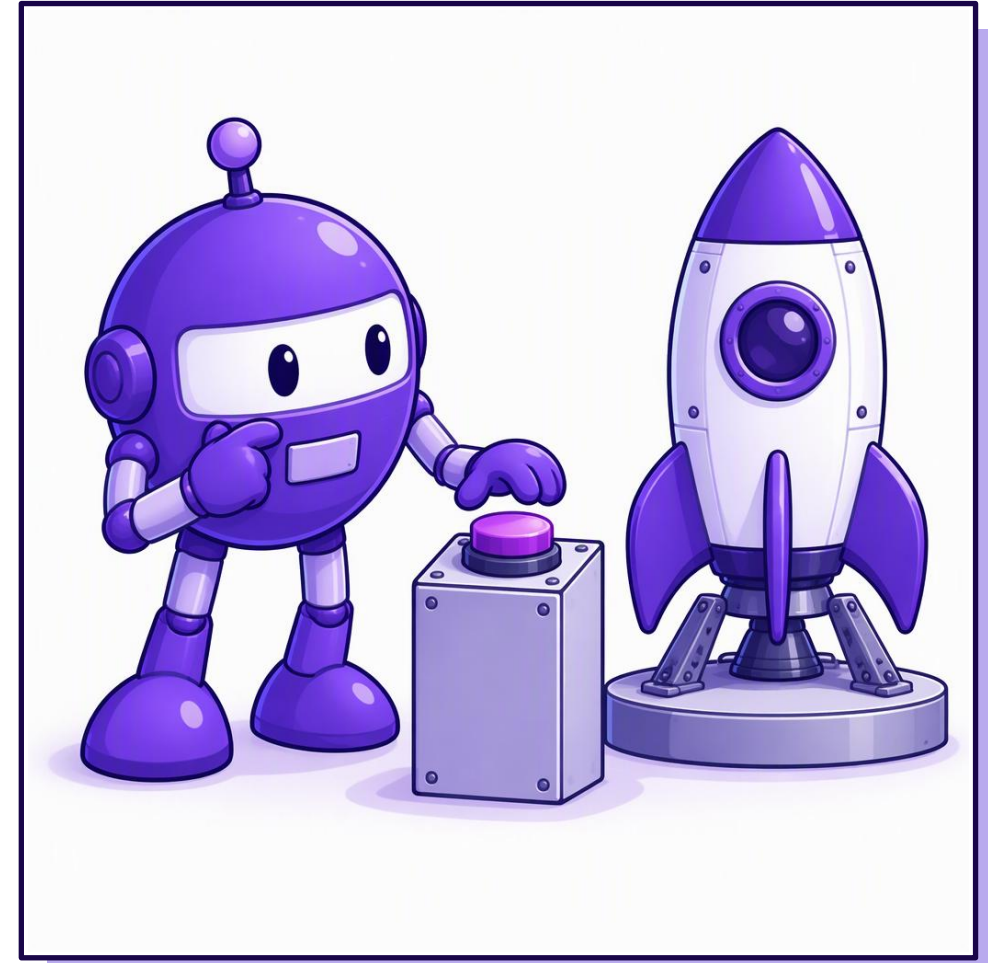


Start with a plain C# sample. Then inspect its MAF bridge.



What would you check before you ship?

The tool call?
The boundary?
The regression?



No surprise deployments. No private data on screen.